Jiaqi Kevin Yang

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EDUCATION

MSc in Computer Science at McGill University **Sep 2026 - Dec 2027**

BSc in Software Engineering at McGill University **Sep 2022 - May 2026**

- Cumulative GPA: 3.74/4.00
- Student exchange at Hong Kong Polytechnic University

EXPERIENCE

Teaching Assistant at McGill University **Aug 2026 - Present**

- Mentored **200+** students in digital circuits, MIPS assembly and computer architecture fundamentals
- Evaluated student assignments to give targeted code reviews and technical feedback

QA Automation Intern at dormakaba **May 2025 - Aug 2025**

- Built a custom Locust load testing framework simulating up to **1000** concurrent users in a smart lock system
- Improved hardware integration tests using Raspberry Pi, reducing manual testing by **~20** hours per release
- Integrated Flask-based internal API and Java-based E2E validation tests into CI/CD pipelines
- Implemented secure, dynamic secret management across **3** repositories using AWS SSO and Parameter Store

PROJECTS

C Compiler **Jan 2026 - May 2026**

- Built a compiler in Java that translated the C programming language into MIPS assembly
- Refactored C grammar to remove loops and developed a recursive-descent parser to generate syntax trees
- Engineered multi-pass semantic analysis engine to enforce strong typing, nested scoping and struct alignment
- Implemented a code generator that allocate registers with heuristic spilling to memory

Smart Dumbbell **Nov 2025**

- Engineered a non-blocking, multithreaded Flask backend to process concurrent MQTT hardware streams, a computer vision pipeline, and real-time workout analytics
- Served as a technical lead for backend and systems integration, coordinating hardware and software teams

Ataxx Game Playing AI **Nov 2025 - Dec 2025**

- Engineered an adversarial game AI using minimax search with alpha-beta pruning and iterative deepening
- Implemented root level heuristic move sorting, cutting depth-4 search latency by nearly **50%**
- Reduced search-tree branching by **33%** by applying move folding logic that pruned redundant state transitions

Distributed Middleware System **Sep 2025 - Oct 2025**

- Built a middleware server in Java to handle concurrent requests and route them to replicated resource servers
- Developed a non-blocking protocol using TCP Sockets, enabling async messages to ensure high throughput
- Designed remote atomic procedure calls to synchronize shared-memory states without data races

SKILLS

Programming Languages: Java, Python, C#, C, C++, Bash, SQL, GraphQL, JavaScript/TypeScript, HTML
Technical Skills: Git, Agile, Linux/Unix, Docker, REST APIs, Django, AWS, Kubernetes, PyTorch, Cursor
CS Fundamentals: Operating Systems, Distributed Systems, Concurrent Programming, Machine Learning
Languages: Fluent English, Fluent French, Advanced Mandarin Chinese, Beginner Japanese